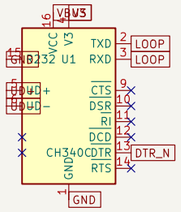


solder pads for USB-A pigtail to the Pi
(VBUS, D-, D+, GND, shield->GND)



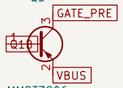
USB-serial bridge; only DTR# is used
TXD->RXD loopback lets the server
probe that the dongle is present



Q1 base current limit



holds Q1 off while DTR# floats
during USB enumeration



polarity inverter: DTR# is active-low,
idles high -> idle/unplugged = radio OFF



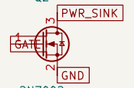
with C4: ~0.3s turn-on delay



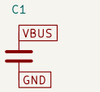
gate glitch filter: with R2/R3 swallows
DTR pulses from USB enumeration
(~0.3s on, ~1s release)



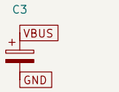
gate pulldown: guarantees OFF when
unpowered; ~1s release with C4



open-collector sink for ACC pin 8:
only ever pulls low, never sources



CH340C VCC decoupling
(place at pin 16)



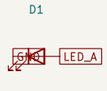
VBUS bulk reservoir
for pigtail inductance



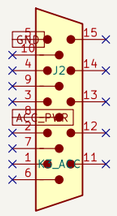
V3 internal-regulator bypass
(required by CH340 datasheet)



power LED current limit (~3mA)



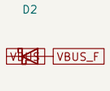
USB power indicator



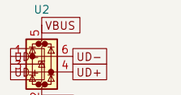
K3 ACC: pin 8 = PWR ON (sink = on),
pin 5 = GND, all others unused



protects the Pi USB port from
board or hood-wiring faults



reverse-polarity guard for a
miswired pigtail (~0.3V drop)



IEC 61000-4-2 ESD clamp
for D+/D-/VBUS



clamps ESD/spikes on the ACC line;
mount at connector, cathode to line



limits fault current into ACC wiring;
negligible vs the K3 pull-up





Fail-safe: unplugged/unpowered/port-closed = radio OFF
CH340C DTR -> PNP inverter -> RC -> 2N7002 open collector -> K3 ACC pin 8
W6JY

Sheet: /
File: k3pwr-dongle.kicad_sch

Title: k3pwr USB power dongle

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/1